LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Daily Practice: Ch 3 Metals and Non-metals

Chapter / Topic: Metals and Non-metals · Week 2 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. One day at a time, about 25 minutes each: Day 1 on the day you study the chapter, then Day 2 and Day 3. Write answers in your notebook and check them with the key the same day. Copy every mistake into your error log.
2. Keep the reactivity series in front of you on Day 1. By Day 3 you should know it by heart: K Na Ca Mg Al Zn Fe Pb [H] Cu Hg Ag Au.

DAY 1: PROPERTIES OF METALS AND NON-METALS

Q1. Which of these non-metals is lustrous (shiny)?

(a) sulphur (b) phosphorus
(c) iodine (d) nitrogen

Q2. Which metal will melt if you hold it in your palm?

(a) sodium (b) mercury
(c) lead (d) gallium

Q3. Which non-metal is a good conductor of electricity?

(a) sulphur (b) phosphorus
(c) iodine (d) graphite (a form of carbon)

Q4. Which of these metals does not react with water at all, not even with steam?

(a) sodium (b) copper
(c) calcium (d) magnesium

Q5. Give reasons: (a) Cooking vessels are made of metals. (b) Electric wires are made of copper and covered with PVC or rubber.

Q6. What do you see when (i) a small piece of potassium is dropped into cold water and (ii) calcium is put into water? Write the balanced equations, and say why calcium starts floating.

Q7. Write balanced equations for the reactions of (i) magnesium with hot water and (ii) aluminium with steam. Name one metal that does not react even with steam.

Q8. What is an alloy? What is an amalgam? Name the metals in brass and in stainless steel.

DAY 2: REACTIVITY SERIES AND IONIC COMPOUNDS

Q9. Which of these reactions will NOT take place?

(a) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (b) $Zn + FeSO_4 \rightarrow ZnSO_4 + Fe$
(c) $2Ag + CuSO_4 \rightarrow Ag_2SO_4 + Cu$ (d) $Mg + ZnSO_4 \rightarrow MgSO_4 + Zn$

Q10. Aqua regia, which can dissolve gold, is a mixture of:

(a) conc. HCl and conc. HNO_3 in the ratio 3 : 1
(b) conc. HCl and conc. HNO_3 in the ratio 1 : 3
(c) conc. H_2SO_4 and conc. HNO_3 in the ratio 3 : 1
(d) conc. HCl and conc. H_2SO_4 in the ratio 1 : 1

Q11. The electronic configuration of the sodium ion, Na^+, is:

(a) 2, 8, 1 (b) 2, 8
(c) 2, 8, 2 (d) 2, 7

Q12. Name two metals that are found free (uncombined) in nature. Why are they found this way?

Q13. A yellow non-metal X burns in air with a blue flame. Its oxide turns moist blue litmus red. Identify X and its oxide, and write the balanced equations for burning X and for its oxide dissolving in water.

Q14. Metals A, B, C and D were added to solutions of $FeSO_4$, $CuSO_4$, $ZnSO_4$ and $AgNO_3$. A displaces copper but not iron. B displaces iron but not zinc. C displaces only silver. D displaces none of them.
(i) Which metal is the most reactive? (ii) What will you see if B is added to copper sulphate solution? (iii) Arrange A, B, C and D in decreasing order of reactivity.

Q15. Give reasons:
(a) Silver articles turn black after some time in air.
(b) Sodium and potassium catch fire in water, but magnesium does not.
(c) Lemon or tamarind juice is used to clean tarnished copper vessels.

DAY 3: EXTRACTION, CORROSION AND CHALLENGE

Q16. **Assertion (A):** In the electrolytic refining of copper, a thick block of impure copper is made the anode.
Reason (R): At the anode, copper atoms of the impure metal lose electrons and go into the solution as ions.
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q17. Which ore is converted into its oxide by calcination (heating strongly in limited air)?
(a) zinc blende, ZnS
(b) calamine, $ZnCO_3$
(c) cinnabar, HgS
(d) copper glance, Cu_2S

Q18. Which of these metals is obtained by the electrolysis of its molten chloride?
(a) iron
(b) copper
(c) sodium
(d) zinc

Q19. In Neha's kitchen, leftover food is wrapped in aluminium foil, the steel cooking pots have copper bottoms, and the pot handles are made of plastic or wood.

(i) Which property of aluminium lets it be rolled into thin foil? Why doesn't the foil react with food quickly?

(ii) Why do cooking pots have copper bottoms?

(iii) Why are the handles made of plastic or wood and not of metal?

Q20. Manganese is obtained by heating manganese dioxide with aluminium powder:
$3MnO_2 + 4Al \rightarrow 3Mn + 2Al_2O_3 + \text{heat}$
(a) Which substance is oxidised and which is reduced? (b) Why is aluminium able to do this? (c) Why is a lot of heat given out, and why is that useful?

Q21. (a) Draw the electron-dot structures of sodium, oxygen and magnesium.
(b) Show the formation of Na_2O and MgO by the transfer of electrons.
(c) Name the ions present in each compound.

~ If you can say the reactivity series without looking, you've won half this chapter! ~